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Dr. Chris Carnie
Klinik Kinderheilkunde III (Onkologie, Hämatologie und Immunologie, Pneumologie)
Hopp Children's Cancer Center Heidelberg (KiTZ) / Universitätsklinikum Heidelberg
Im Neuenheimer Feld 350, 69120 Heidelberg
Job-ID: V000015650

Re: PhD position — Genome Stability and Nucleotide Metabolism in Cancer (V000015650)

Dear Dr. Carnie,

I am applying for the doctoral position in your group on the Emmy Noether-funded project "Nucleotide Metabolism and Genome Stability in Cancer," and I am grateful to you for drawing my attention to it. The project sits very close to where I trained: my research began on chemically-induced DNA damage, and my work since has been computational. Because the position spans wet lab, dry lab, and bioinformatic analysis of the 'omics data the projects generate, I am applying as someone who can contribute across that whole span rather than at one end of it.

The biology is where I began. My bachelor's thesis was an experimental study of DNA interstrand crosslinking — the formation of crosslinks and the cellular response to crosslinking agents. That is directly adjacent to the antimetabolite biology at the centre of your project: both crosslinkers and antimetabolite drugs are agents that perturb the cell and impose potentially cytotoxic DNA damage, studied through the same repertoire of cell-survival, repair, and damage-response assays. I have a molecular and cell-biology grounding (BSc in Biochemistry and Cell Biology, MSc in Pharmaceutical Biotechnology), bench training including mammalian cell culture and microscopy, and a genuine wish to return to laboratory work in exactly this area — the interplay between a metabolic perturbation, the DNA damage it causes, and the pathways that determine whether a cancer cell survives it.

The analysis is my strongest contribution. The position lists bioinformatic aptitude with R or Python as desirable, and the tasks include dry-lab investigation and analysis of the 'omics data you generate. This is where I am strongest. My doctoral work was at TUM's Chair of Proteomics and Bioanalytics, so mass-spectrometry and proteomic / metabolomic data analysis is home ground for me — directly relevant to the metabolomics in your toolkit. I have built a multi-omics inference pipeline integrating genotype, transcriptomics, and metagenomics, validated against 1000 Genomes, gnomAD, and UK Biobank. Genome-wide CRISPR screens, at the analysis layer, are a genetic-interaction problem — identifying the dependencies that emerge only against a metabolic or repair-deficient background — and I would be glad to own that analysis for the group while contributing at the bench. A candidate who can both generate the data and analyse it is, I think, what makes the dual wet/dry structure of these projects work.

Relevant computational work. My independent research has been in computational oncology, drug metabolism, and pharmacology: tumour characterisation, neoantigen and MHC-binding prediction (AUC 0.81 on 2,847 IEDB peptides), and pharmacokinetic / pharmacodynamic and drug-metabolism modelling validated against NIST reference data — the last of these directly concerned with how antimetabolite-class compounds are processed and act. These tools are openly available (linked above) should they be of interest.

Background. I hold an MSc in Computational Biology (Universität Tübingen), an MSc in Pharmaceutical Biotechnology (HAW Hamburg), and a BSc in Biochemistry and Cell Biology (Jacobs University Bremen). I conducted doctoral research in Computational Lipidomics at TUM (2019–2021) before transitioning to independent research. I have lived and worked in Germany continuously since 2011; English is native and my German is conversational, which suits an English-language working environment well.

I am available at the earliest possible date and would welcome the opportunity to discuss how this combination of experimental DNA-damage background and computational analysis could contribute to your team.

Yours sincerely,

Kundai Farai Sachikonye